

Brainstorming & Idea Prioritization Template

Date	03 July 2025
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	3 Marks

Step 1: Brainstorm and Idea Listing

Each team member lists out as many ideas as possible without judging them at this stage.

S.No	Team Member	Idea / Suggestion	Category	Group No.
1	Avadhi Kukkamalla	Merge application_record.csv and credit_record.csv using applicant ID to create unified dataset	Data Collection	1
2	Avadhi Kukkamalla	Convert multi-class STATUS codes (0-5, C, X) into binary risk label: 1=High Risk, 0=Low Risk	Feature Engineering	1
3	Avadhi Kukkamalla	Handle missing OCCUPATION_TYPE values (134,203 records) by filling with 'Unknown' category	Data Pre-processing	2
4	Avadhi Kukkamalla	Create EDA visualizations: income distribution, age chart, risk rate by gender and education type	Data Analysis	2
5	Avadhi Kukkamalla	Apply class_weight='balanced' to handle 88%/12% class imbalance during model training	Model Building	3
6	Avadhi Kukkamalla	Save trained XGBoost model as .pkl file and load it in Flask app for real-time predictions	Application Building	4
7	Ram Charan Teja Naramsetty	Engineer AGE_YEARS and YEARS_EMPLOYED features from raw negative day values in dataset	Feature Engineering	1
8	Ram Charan Teja Naramsetty	Train and compare four ML models: Logistic Regression, Decision Tree, Random Forest, XGBoost	Model Building	3

S.No	Team Member	Idea / Suggestion	Category	Group No.
9	Ram Charan Teja Naramsetty	One-hot encode categorical features (income type, education, occupation) for model input columns	Data Pre-processing	2
10	Ram Charan Teja Naramsetty	Build Flask web app with single prediction form and batch CSV screening for compliance officers	Application Building	4
11	Ram Charan Teja Naramsetty	Create model comparison bar chart showing accuracy, precision, recall and F1 for all four models	Data Analysis	2
12	Ram Charan Teja Naramsetty	Deploy best-performing XGBoost model to IBM Watson Machine Learning cloud platform pipeline	Cloud Deployment	5

Step 2: Idea Prioritization

Rate each grouped idea on feasibility and importance, then select the final idea(s) to move forward with.

Group No.	Final Idea	Feasibility (High/Medium/Low)	Importance (High/Medium/Low)	Priority	Selected (Yes/No)
1	Data Collection & Feature Engineering (Merge datasets, build binary risk label, engineer age & employment features)	High	High	1st	Yes
2	Data Pre-processing & Analysis (Handle missing values, encode categoricals, create EDA visualizations & reports)	High	High	2nd	Yes
3	Model Building & Evaluation (Train 4 ML models, handle class imbalance, compare accuracy/recall/F1 metrics)	High	High	1st	Yes
4	Application Building (Flask web app: single prediction form + batch CSV upload for compliance screening)	High	High	2nd	Yes
5	Cloud Deployment (IBM Watson Machine Learning pipeline for scalable real-time cloud predictions)	Medium	High	3rd	Yes